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Selected Academic and Technical Work

Structural Engineering Portfolio

This portfolio presents a selected set of academic and technical work developed during my studies in Structural Engineering at Politecnico di Milano, with a focus on structural analysis, seismic design, wind engineering, and BIM based modelling.

Profile

Recent MSc graduate in Structural Engineering from Politecnico di Milano, with a strong interest in structural design, dynamic behaviour of structures, and integrated design processes. My experience combines analytical work, numerical modelling, seismic design, and BIM based workflows.

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A Computational Framework for Wind Induced Serviceability of Tall Buildings

Frequency domain workflow, response to buffeting and VIV

MSc Thesis, Structural Engineering, Politecnico di Milano
Author: Fabio Menconi
Supervisors: Prof. Francesco Foti, Co supervisor: Prof. Federico Perotti, Bruno Dal Lago
Year: 2026

Context and goal

In tall buildings, wind loading often governs serviceability rather than strength, especially when occupant comfort, inter storey drift and facade related demands are considered. In this context, designers need tools that can provide a reliable first estimate of the dynamic response before moving to more expensive and time consuming investigations. The aim of this thesis is therefore to develop a practical frequency domain workflow able to predict the main wind induced response quantities, while also supporting early decisions on possible response mitigation strategies, such as changes in stiffness, mass distribution or damping.

Method in brief

The workflow evaluates the wind induced response in the frequency domain by combining structural modal properties with aerodynamic loading models and spectral descriptions of wind fluctuations. The procedure starts from the modal characteristics of the building, which are used to define the dynamic amplification through the transfer functions, and then combines them with spectral load models for buffeting, by Solari and Piccardo (2001), and vortex induced vibration, by Liang (2002). In this way, both broadband turbulence driven effects and narrow band resonant response can be represented, leading to estimates of displacement and acceleration spectra, as well as the corresponding RMS and peak values.

Code architecture snapshot

Inputs geometry, site and exposure parameters for $U(z)$, $I_u(z)$, $L_u(z)$, modal shapes and frequencies, damping, angle of attack set

Core ESDU based aerodynamic coefficients, buffeting and VIV spectra, frequency integration, modal superposition

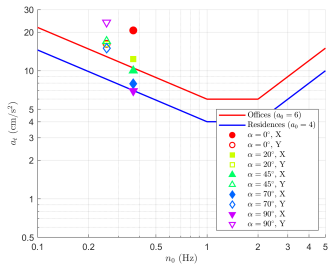
Outputs $S_a(f)$, RMS and peak accelerations, drift ratio envelopes $\theta(z)$, base shear, base bending and torsional moments split into mean, dynamic and total components, comfort checks and CNR DT 207 comparison, automated export

Validation and case studies

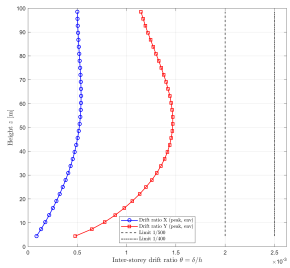
Benchmarks include the CAARC A and B buildings, used to assess peak accelerations and base actions, and the spectral response $S_a(f)$ of the Polimi aeroelastic model studied by Cavenaghi. These comparisons show good agreement, both in terms of response level and its variation with the angle of attack. The same procedure is applied to a realistic precast tower case study.

Representative results

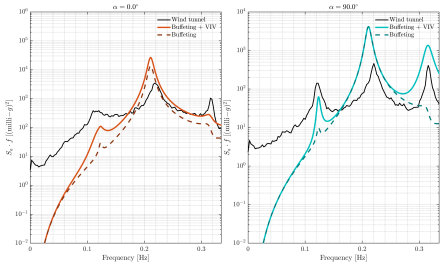
The results highlight the complementary role of the two response mechanisms. Buffeting reproduces the broadband background of the response, while the inclusion of vortex induced vibration is necessary to capture the narrow band resonant peaks that may dominate in cross wind conditions.



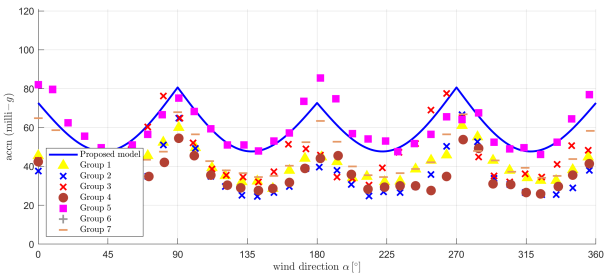
Precast, acc. comfort check



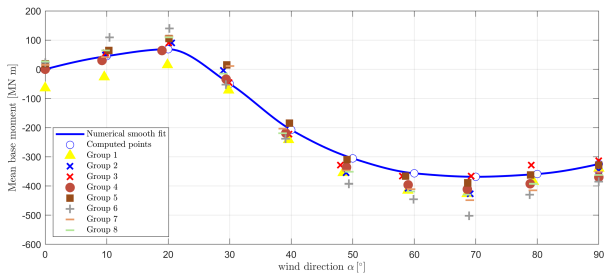
Precast, drift ratio envelope



Polimi model, $S_{ax}(f)$, log scale



CAARC B, peak corner roof acc. vs α



CAARC A, mean base moment M_y vs α

Key takeaway The workflow helps identify the factors that govern wind induced serviceability, including the relative influence of mean and fluctuating contributions, structural stiffness, damping and building orientation. It also provides a practical benchmark for comparison with wind tunnel and CFD based results.

Structural Engineering Internship

BMS Progetti, Milan

Professional experience in FEM modelling, load assessment, and reinforced concrete verification for the M5 Cinisello-Monza underground station

Year: 2025

During this internship I worked on the structural verification of a large underground reinforced concrete infrastructure developed in a professional design environment. My activity focused on FEM modelling support in MIDAS Gen, definition and interpretation of load cases and combinations, reading of global and local response, and checking of structural elements such as slabs, walls, beams, columns, and foundation members under static, seismic, hydraulic, and soil related actions.

Scope of the experience

The internship gave me direct exposure to the verification workflow of a real underground infrastructure project, where structural behaviour had to be assessed in relation to construction logic, soil interaction, permanent and variable loads, and code based safety checks. A central aspect of the experience was understanding how complex actions are translated into a reliable numerical model and then interpreted for design verification.

Main activities

My contribution included support to the FEM modelling of reinforced concrete structural components in MIDAS Gen, using beam and plate idealisations, elastic soil restraints, and load application consistent with the structural scheme. I reviewed permanent and variable loads acting on underground station elements, including self weight, floor loads, track related actions, backfill and cover loads, machinery and plant room overloads, escalator and lift related loads, water pressure, temperature effects, and load redistribution from surface traffic. I also worked on the interpretation of load combinations at ULS and SLS, including seismic combinations for underground structures, and on the reading of envelopes of axial force, bending moments, shear forces, reactions, and local plate actions for the verification of slabs, walls, beams, columns, and foundation elements.

What I learned

This experience helped me understand how professional structural verification differs from academic modelling. I became more familiar with the practical use of MIDAS Gen for large three dimensional models, with the interpretation of code based actions rather than only their theoretical definition, and with the importance of keeping modelling assumptions, loading logic, and documentation fully aligned. It also gave me a clearer understanding of underground structural behaviour, especially the role of soil restraints, paratia induced actions, hydrostatic effects and seismic input for buried structures.

Technical highlights

Context Structural engineering internship
Project M5 Cinisello-Monza underground station
Type of project Underground reinforced concrete infrastructure
Main software MIDAS Gen, Excel
MIDAS functions used 3D FEM modelling, beam and plate elements, elastic soil springs, static and seismic load cases, load combinations, envelopes, reactions, internal force diagrams, plate stress resultants
Main actions considered Self weight, permanent floor loads, track loads, backfill, machinery loads, escalators, lift areas, hydrostatic pressure, temperature, traffic surcharge, seismic actions
Verification topics RC slabs, walls, beams, columns, foundations, serviceability and ultimate checks, uplift assessment, interpretation of global and local response

Confidentiality note

This page intentionally presents the internship in a general and non project sensitive form. Detailed calculation outputs, drawings, internal documents, and proprietary information are not included for confidentiality reasons.

Key takeaway This internship was my first direct experience of structural verification for a real underground project, where the quality of the engineering work depended not only on obtaining results, but on building a coherent model, applying loads correctly, interpreting combinations carefully, and reading the structural response.

Seismic Design and Assessment Projects

Three complementary academic projects in modelling, design, and vulnerability assessment

Academic projects, Earthquake Resistant Design, Politecnico di Milano Year: 2025

These three projects allowed me to work on seismic engineering from different perspectives: numerical modelling and calibration against experimental results, design of a new reinforced concrete building according to Eurocode 8, and simplified vulnerability assessment of an existing masonry structure. Together, they helped me build a more complete understanding of structural response, code based design and the role of modelling assumptions.

1. Dynamic modelling of an RC frame

This project focused on the numerical modelling of the ELSA reinforced concrete dual frame-wall structure under both low and high seismic excitation. The work was developed in MIDAS Gen by progressively refining modelling assumptions in order to improve agreement with experimental modal properties and observed response. The final model was then used for response spectrum, time history, and pushover analyses, with comparison against experimental displacements, base shear, and capacity curves.

Focus: MIDAS Gen, model calibration, elastic and inelastic analysis, comparison with experimental data

2. Design of a new RC building

This project involved the seismic design of a six storey reinforced concrete residential building located in Assisi, developed according to Eurocode 8 for a DCH structure. The work included structural layout definition, load analysis, regularity checks, 3D modelling in MIDAS Gen, modal response spectrum analysis, and reinforcement design of beams, columns, and walls. Particular attention was given to ductility requirements, drift verification, and the contribution of walls to the global behaviour and torsional stability of the building.

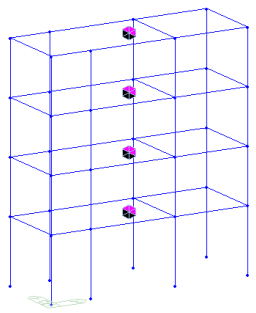
Focus: Eurocode 8, 3D modelling, RC design, ductility and drift checks

3. Vulnerability assessment of masonry

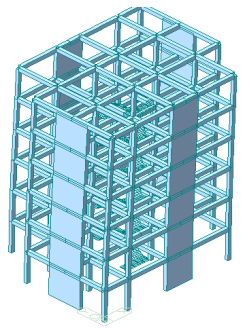
This project dealt with the simplified seismic vulnerability assessment of an existing three storey masonry building located in L'Aquila, using the LV1 procedure from NTC2018 and the Italian Guidelines. The work included load evaluation, computation of wall areas and masses, irregularity coefficients, shear capacity, spectral acceleration, and safety related indicators. It was particularly useful for understanding how simplified assessment procedures still require careful interpretation of geometry, materials, structural layout, and global response.

Focus: existing structures, masonry, LV1 procedure, vulnerability indicators

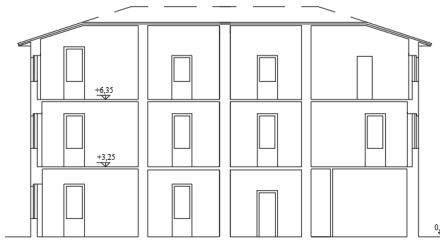
Selected material



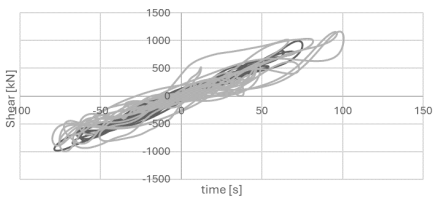
ELSA frame, diaphragm behaviour



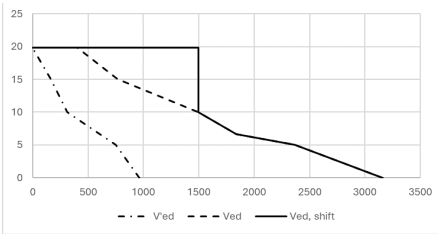
Assisi RC building, structural model



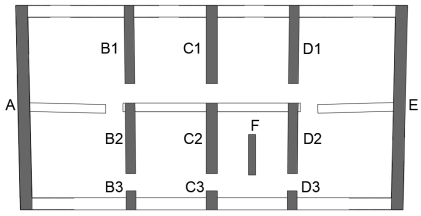
Masonry building, view



ELSA frame, Base shear vs Displacement



Assisi RC building, Shear envelope



Masonry building, Walls y-direction

Key takeaway Taken together, these projects gave me experience with three complementary dimensions of seismic engineering: model calibration and interpretation, code based design of new structures, and assessment of existing buildings under simplified procedures.

Prestressed Box Girder Bridge Design

Single span concrete bridge, longitudinal and transversal design

Academic project, Bridge Theory and Design, Politecnico di Milano Year: 2025

This project focused on the design of a single span prestressed concrete box girder bridge for a C1 suburban secondary road. The work combined global longitudinal design, prestressing layout, serviceability and ultimate verifications, and detailed transversal analysis of the box section. A relevant part of the project was the integration of code based checks with practical design decisions on geometry, tendon arrangement, anchorage zones, and reinforcement detailing.

Technical highlights

Type Prestressed concrete box girder bridge
Scheme Single span, simply supported
Span 36.00 m
Deck width 12.00 m
Section depth 2.10 m
Road category C1 suburban secondary road
Materials C35/45, B450C, Y1860S7
Main topics Prestressing layout, SLS and ULS checks, shear and torsion, transverse bending, anchorage detailing

Longitudinal design

The longitudinal design included permanent and traffic loads according to Eurocode load models, serviceability and ultimate combinations, and identification of the load positions that maximize bending moment and shear. Prestressing was introduced through post-tensioned tendons placed in both webs. The tendon layout was developed in the parallel and perpendicular planes, with explicit control of cover, spacing, anchorage geometry, and deviation of the cable profiles between end section and midspan. Prestressing losses were evaluated analytically, separating immediate losses, due to friction and wedge draw-in, from time dependent losses, and were then used for the stress checks at different construction and service stages.

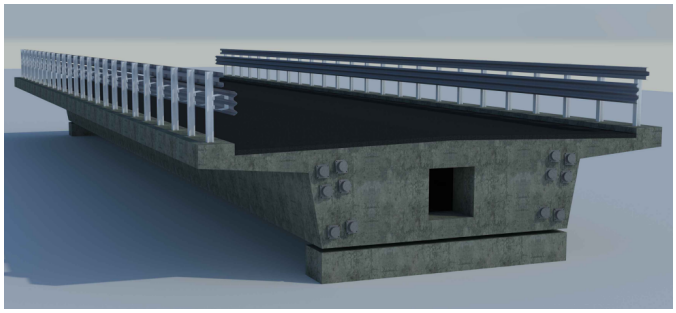
Transversal analysis and detailing

The project also included a detailed transverse analysis of the box girder section. Distributed transverse loads and concentrated loads were studied through sectional models and Homberg surfaces, leading to the final bending moment envelopes used for reinforcement design. Longitudinal and transverse reinforcement were then defined together with anchorage lengths and local reinforcement in the anchorage zone through a strut and tie interpretation of the support region.

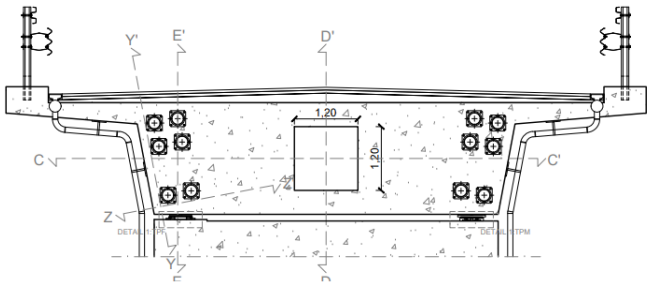
My contribution

Geometry definition, load analysis, tendon layout, prestressing loss evaluation, SLS and ULS verifications, transverse analysis, reinforcement detailing, and preparation of final calculation outputs and drawings.

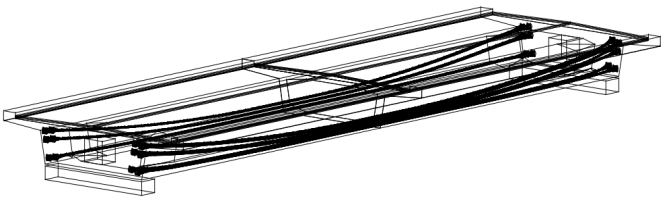
Selected material



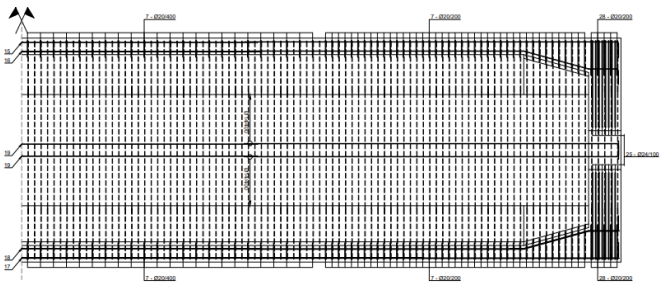
Rendered prospettic view



Anchorage zone section



Cable layout, wireframe view



Stirrups detailing, longitudinal section

Country House Project in Revit

Federated BIM workflow for modelling, coordination, and documentation

Academic project, BIM Fundamentals and Applications, Politecnico di Milano Year: 2025

This project focused on the Revit based development of a two storey country house with a nearby garage, moving from conceptual design to structural modelling, architectural refinement, and final federated coordination. The work was not limited to geometric modelling, but included model organisation, parametric control, coordination between linked models, schedules, interference checking, quantity take off, cost evaluation, and final sheet composition.

Project overview

The project started from a conceptual Revit model used to define grids, levels, main reference objects, roof geometries, and the overall spatial organisation of the building. This initial stage was then developed into a structural model through linked references, copied levels and grids, structural framing, floor and roof elements, foundations, sample reinforcement, and analytical model adjustments. In parallel, the architectural model was refined through material layering, curtain walls, handrails, stairs, topography, and custom families, progressively increasing the level of detail and consistency of the project.

Workflow and coordination

A relevant part of the exercise was the coordination between conceptual, structural, and architectural models. The workflow included Revit linking, Copy/Monitor operations, coordination review after changes in the linked model, and the final construction of a federated model. Particular attention was given to interference checks between structural and architectural elements, with correction procedures based on join and cut operations. This made the project especially useful for understanding BIM not only as modelling, but as a coordinated design environment in which geometry, information, and consistency have to be managed together.

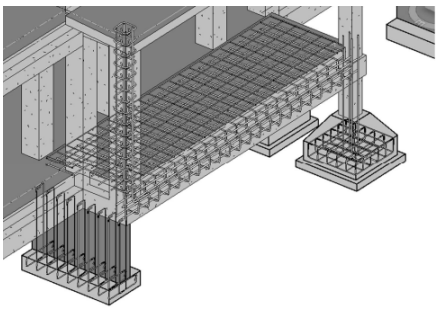
Technical highlights

Building Two storey country house
Main software Autodesk Revit
Main topics Conceptual modelling, structural modelling, architectural refinement, coordination, schedules, sheets
Key operations Copy/Monitor, linked models, custom families, interference check, material take off, cost computation
Outputs Federated model, custom parametric objects, schedules, renders, final sheets

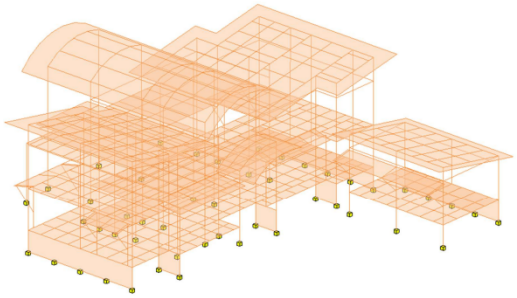
Information modelling and outputs

The final phase of the work involved the production of schedules and documentation directly from the model. This included window and room schedules, material take off schedules for brick and concrete, cost computation through calculated parameters, construction phases, and the creation of final sheets combining plans, sections, elevations, schedules, and renders. The project also included the creation of custom parametric families, in particular a door with side windows and a dedicated handle family, to test parametric control and nested family logic.

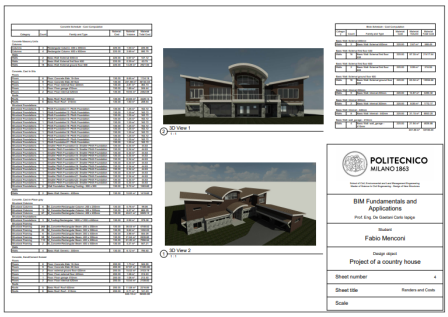
Selected material



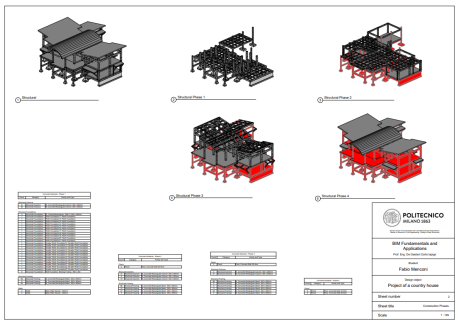
Reinforcements of structural elements



Analytical model



Renders and costs board



Construction phases board